Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2017-I-Q03] | Part A (Q1-Q5)

Answer: 069

1. Triangular numbers at indices $n+20, n$ differ by $20n+210$, divisible by10. Thus the last digits have period20.

2. One period is $(1,3,6,0,5,1,8,6,5,5,6,8,1,5,0,6,3,1,0,0)$, with sum70. Since $2017 = 100 \cdot 20 + 17$, the remaining first17 terms sum69.

3. Total 7069 has remainder69, or AIME format069.

Source / 来源

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2018-II-Q02] | Part A (Q1-Q5)

Answer: 112

1. The initial terms are $2, 5, 8, 5, 6, 10, 7, 4, 7, 6, 2, 5, 8$. The consecutive triple at indices ten through twelve repeats the initial triple.

2. Because three terms determine every future term, this proves period ten. The requested terms are $a_8 = 7, a_0 = 2, a_2 = 8$, whose product is 112.

Source / 来源

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2001-II-Q10] | Part B (Q6-Q10)

Answer: 784

1. Since 10 is coprime to 1001, divisibility is equivalent to $10^{j-i} \equiv 1 \pmod{1001}$. The orders modulo $7, 11, 13$ are $6, 2, 6$, so $j - i$ must be a multiple of 6.

2. Different pairs give different integers, because the number has i trailing zeros followed by exactly $j - i$ nines. For gap $6d$, there are $100 - 6d$ choices, $1 \leq d \leq 16$.

3. The total is $\sum_{d=1}^{16}(100 - 6d) = 784$.

Source / 来源

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2021-II-Q09] | Part B (Q6-Q10)

Answer: 295

1. If an odd prime divides both numbers, the order of 2 modulo that prime has 2-adic valuation exactly $v_2(m)+1$ and divides n. Necessarily $v_2(n) > v_2(m)$.

2. Conversely, under this inequality let $g = \gcd(m,n)$. Then m/g is odd and n/g even, so 2^g+1 divides both numbers. Thus the criterion is sufficient too.

3. The counts of $v_2 = 0,1,2,3,4$ in 1 through 30 are $15, 8, 4, 2, 1$. Therefore the pair count is $15\cdot 15 + 8\cdot 7 + 4\cdot 3 + 2\cdot 1 = 295$.

Source / 来源

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乘法阶、模幂与周期

[AIME-2011-I-Q11] | Part C (Q11-Q15)

Answer: 007

1. The first three residues are $1, 2, 4$; all later ones are divisible by 8 and not by 5, so these classes are disjoint.

2. Modulo 125, $2^{10} = 24, 2^{20} = 76, 2^{40} = 26$, so $2^{50} = 26 \cdot 24 \equiv -1$. Hence for $n \geq 3$, $2^{n+50} \equiv -2^n \pmod{1000}$, using divisibility by 8 as well.

3. The later residue set is therefore closed under $r \mapsto 1000 - r$. There is no fixed point: 0 and 500 cannot be powers of two modulo 1000. Pairing distinct residues makes their total a multiple of 1000.

4. Only $1 + 2 + 4$ remains, so the answer is 7.

Source / 来源

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[AIME-2018-I-Q11] | Part C (Q11-Q15)

Answer: 195

1. The condition is $3^n \equiv 1 \pmod{143^2}$, equivalent to congruence one modulo 121 and 169.

2. Modulo 121, $3^5 = 243 \equiv 1$ and $3 \not\equiv 1$, so the order is five. Modulo thirteen, the order is three; for $n = 3k$, binomial expansion gives $3^{3k} = (1+26)^k \equiv 1 + 26k \pmod{169}$, since 26^2 is divisible by 169.

3. This is one exactly when $13 \mid k$, so the order modulo 169 is thirty-nine. The least simultaneous exponent is $\operatorname{lcm}(5, 39) = 195$.

Source / 来源

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2019-I-Q14] | Part C (Q11-Q15)

Answer: 097

1. If odd prime p divides it, 2019 has multiplicative order exactly sixteen modulo p: its sixteenth power is one, but its eighth is minus one. Hence $p \equiv 1 \pmod{16}$. The first such prime candidates are seventeen and ninety-seven; intermediate numbers $33, 49, 65, 81$ are composite.

2. Modulo seventeen, $2019 \equiv 13$ and $13^2 \equiv -1$, so its eighth power is one, not minus one. Modulo ninety-seven, successive squared residues are $33, 22, -1$, so divisibility holds.

3. The least odd prime factor is 97.

Source / 来源

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2020-I-Q12] | Part C (Q11-Q15)

Answer: 270

1. Because $149 - 2 = 3 \cdot 7^2$, the odd-prime lifting formula gives $v_3 = 1 + v_3(n)$ and $v_7 = 2 + v_7(n)$. Thus $3^2 7^5 \mid n$.

2. Modulo 5, $(149/2)^n \equiv 2^n$, whose order is 4. Write $n = 4k$. Since $v_5(149^4 - 2^4) = 1$, lifting gives $v_5(149^n - 2^n) = 1 + v_5(k)$, so $4 \cdot 5^4 \mid n$.

3. These conditions are necessary and sufficient. The least exponent is $2^2 3^2 5^4 7^5$, which has $(2+1)(2+1)(4+1)(5+1) = 270$ divisors.

Source / 来源

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2023-II-Q15] | Part C (Q11-Q15)

Answer: 363

1. Write $a_n = 1 + k_n 2^n$. Minimality makes k_n the representative in $\{1, \ldots, 22\}$ of $-2^{-n} \pmod{23}$. Equality with a_{n+1} holds exactly when k_n is even.

2. Since $2^{11} \equiv 1 \pmod{23}$ and the order is 11 (a prime, and $2 \not\equiv 1$), the sequence k_n repeats every eleven terms.

3. For $n = 1, \ldots, 11$, its values are $11, 17, 20, 10, 5, 14, 7, 15, 19, 21, 22$. These follow from $k_{n+1} = k_n/2$ for even k_n, or $(k_n + 23)/2$ for odd k_n. Even entries occur at residues $3, 4, 6, 0 \pmod{11}$.

4. In $1, \ldots, 1000 = 90 \cdot 11 + 10$, there are ninety full periods and three extra even entries. The count is $90 \cdot 4 + 3 = 363$.

Source / 来源

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2024-I-Q13] | Part C (Q11-Q15)

Answer: 110

1. $p = 2$ fails modulo 4. For odd p, a root has multiplicative order 8, so $8 \mid p - 1$. The first candidate is 17, whose roots of $x^4 = -1$ are $2, 8, 9, 15$.

2. For each root a, write $m = a + 17t$. Modulo 289, the condition is $(a^4 + 1)/17 + 4a^3 t \equiv 0 \pmod{17}$. The derivative coefficient is nonzero, so each root lifts uniquely.

3. The four lifted residues are $155, 110, 179, 134$, respectively. Therefore the least positive root is 110; indeed $110^4 + 1 = 289 \cdot 506609$.

Source / 来源

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2025-I-Q15] | Part C (Q11-Q15)

Answer: 735

1. Modulo 9, a unit cube is 1 or -1. Thus a solution has exactly two units or no units. Cubing maps the 486 units modulo 3^6 bijectively onto the unit cube classes modulo 3^7, which are precisely those congruent to $\pm 1 \pmod 9$; this follows from the kernel of cubing on units having size 3.

2. With two units, choose the divisible variable in three ways, its value in 243 ways, and one unit in 486 ways. The remaining unit is forced by the bijection. This contributes 354294.

3. If all variables are divisible by 3, divide them by 3 and reduce the cube-sum condition modulo 81. Only their residues modulo 27 matter, with 9^3 lifts. At this lower level the two-unit count is $3 \cdot 18 \cdot 9 = 486$; all-divisible triples reduce to a sum-zero condition modulo 3, giving $9^3/3 = 243$. Thus this case contributes $9^3(486 + 243) = 531441$. Total 885735, remainder 735.

Source / 来源